INTERNATIONAL MONETARY FUND

MALI

August 12, 2019

REQUEST FOR A THREE-YEAR ARRANGEMENT UNDER THE EXTENDED CREDIT FACILITY—DEBT SUSTAINABILITY ANALYSIS

Approved by
Annalisa Fedelino and Maria Gonzalez (IMF) and Marcello Estevão (IDA)

Prepared jointly by the staffs of the International Monetary Fund (IMF) and the International Development Association (IDA)

Mali **Joint Bank-Fund Staff Debt Sustainability Analysis**	
Risk of external debt distress:	Moderate
Overall risk of debt distress	Moderate
Granularity in risk rating	With some space to absorb shocks
Application of judgement	Yes; a customized scenario was used to take account of security challenges and large development needs

Mali remains at moderate risk of external debt distress. This rating is unchanged from the previous analysis and consistent with the May 2018 Staff Report (IMF Country Report/18/141). All the projected external debt burden indicators remain below their thresholds under the baseline. However, the ratio of the external debt service to exports exceeds its threshold in the case of an extreme shock to exports under a customized scenario that incorporates 2 percentage points of GDP larger fiscal deficits over 2019 to 2023 than the baseline.[1]

The baseline scenario assumes improved fiscal policies and achievement of the WAEMU fiscal deficit convergence criteria by 2019. As illustrated in the customized scenario, continued shortfall in domestic revenue mobilization and a deterioration in security conditions would result in a weakened fiscal position and increase the likelihood of debt distress. Mali's main challenge continues to be ensuring macroeconomic stability while protecting social and investment spending and providing for growing security spending and large development needs. To maintain debt at moderate risk rating, it is essential that the authorities continue their efforts to mobilize domestic revenue and implement reforms. Debt management capacity should be strengthened while deepening structural reforms to diversify the exports base.

[1] This DSA was jointly prepared by IMF and World Bank staff under the new debt sustainability framework (DSF) for low-income countries (LICs), implemented since July 2018. The debt-carrying capacity is classified using the country-specific composite indicator (CI) composed of three macroeconomic indicators and the World Bank's Country Policy and Institutional Assessment (CPIA). Mali's capacity is assessed as "medium" using the CI based on the April 2019 WEO and the 2017 CPIA data.

BACKGROUND

A. Public Debt Coverage

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	
7	Central bank (borrowed on behalf of the government)	
8	Non-guaranteed SOE debt	

1. Mali's public debt covers external and domestic obligations of the central government. Information on state-owned enterprises (SOEs) debt is generally not available, and only the liabilities of the national state-owned electricity company EDM (the main SOE), estimated at CFAF 319 billion (3.1 percent of GDP) at end-June 2018, are included in the baseline. Contingent liabilities estimated at CFAF 150 billion (1.5 percent of GDP) from EDM have already been assumed by the central government as a result of EDM's financial difficulties. State and local government entities do not borrow directly on their own. While liabilities of other public sector entities, including SOEs and other guaranteed debts, are not included in the baseline due to data constraints, a contingent liability test is performed to capture potential fiscal risk posed by these entities (Text Table 1). Staff will continue discussions with the authorities to broaden the coverage of public debt to include guarantees and SOE debt. The component of the contingent liability shock related to SOEs that are not accounted for in the public sector coverage is calibrated at 2 percent of GDP (the default setting) (Text Table 1). Public Private Partnerships (PPPs) capital stock for Mali is estimated at 3.7 percent of GDP, implying a calibrated 1.3 percent of GDP (0.35*3.7) for the PPP component of the contingent liability stress test. External debt is defined on a currency-denominated basis.

Text Table 1. Coverage of Public Debt and Size of Contingent Liability Shock

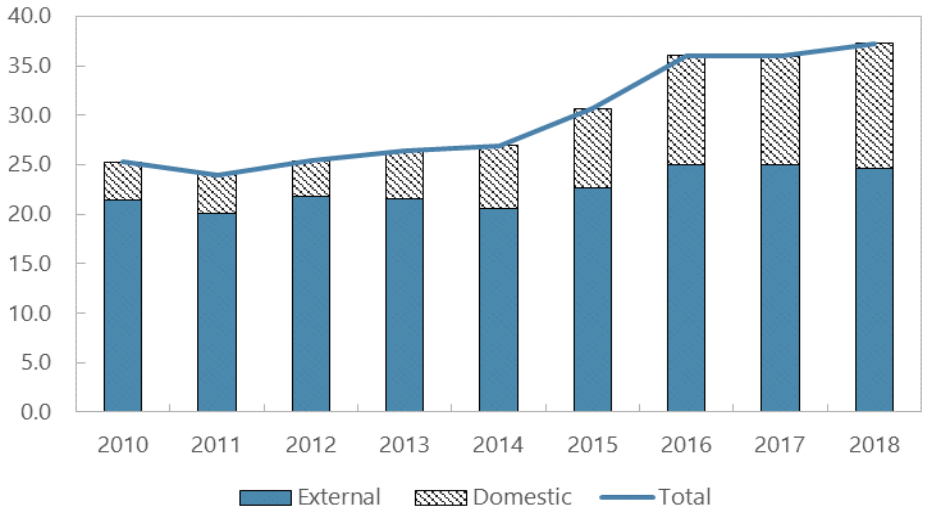
1	The country's coverage of public debt	The central government		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	0.0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	2.0	
4	PPP	35 percent of PPP stock	1.3	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0	
	Total (2+3+4+5) (in percent of GDP)		8.3	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

B. Background on Debt

2. At end-2018, Mali's stock of public debt amounted to CFAF 3,555 billion (37.3 percent of GDP), composed mostly of external debt on concessional terms (Text Figure 1 and Text Table 2). External debt amounted to CFAF 2,346 billion, of which CFAF 1,886 billion is due to multilateral creditors and CFAF 460 billion to bilateral creditors. However, domestic debt is rapidly building up, increasing from a low base of 6.3 percent of GDP in 2014 to 12.7 percent of GDP in 2018, mostly in treasury bills and bonds issued on the WAEMU regional market (Text Table 3).

Text Figure 1. Mali: Central Government Debt 2010–18
(in percent of GDP)



Sources: Malian authorities; IMF Staff estimates.

3. The authorities are planning to use a part of the IDA allocation in 2019 to reprofile domestic debt and reduce rollover risk. The government is planning a debt management operation that will entail borrowing a syndicated loan from international banks guaranteed by the World Bank using part of the 2019 IDA allocation. The authorities view this operation as a precursor for eventually accessing international financial markets and issuing a Eurobond. They intend to use part of the proceed of this operation (up to US$500 million under the Policy-Based Guarantee financial instrument, PBG) to support EDM, reducing its short-term exposure towards the banking sector, and to reduce rollover risk by replacing part of domestic and regional debt with external loans. However, as its timing and size are not yet confirmed, this operation has not been included in the baseline for the new program. Adjusters to performance criteria have been included should the operation materialize before end-2019.

Text Table 2. Mali: External Debt Stock at Year-End, 2010–18
(billions of CFAF)

	2010	2011	2012	2013	2014	2015	2016	2017	2018
Total	1,134	1,229	1,382	1,445	1,463	1,754	2,081	2,228	2,347
(percent of GDP)	21.4	20.1	21.8	22.1	20.6	22.6	25.1	24.9	24.6
Multilateral	896	1,006	1,105	1,160	1,181	1,384	1,634	1,799	1,886
IMF [1]	72	101	101	83	72	80	105	114	179
World Bank/IDA	414	494	578	586	597	728	823	908	947
African Development Bank	158	257	247	229					
Islamic Development Bank	114	124	118	111	92	116	141	118	99
Others	139	30	62	151	419	460	565	660	662
Official Bilateral	236	222	276	284	282	370	447	429	460
Paris Club official debt	10	13	13	10	8	41	41	40	44
Non-Paris Club official debt	226	209	263	275	273	329	406	389	417
Other Creditors	3	2	2	1	1	0	0	0	0

Sources: Malian authorities; and IMF staff estimates.

[1] Includes August 2009 SDR allocation.

Text Table 3. Mali: Public Domestic Debt Stock at Year-End, 2014–18
(billions of CFAF)

	2014	2015	2016	2017	2018
Total	445	622	913	983	1,209
(percent of GDP)	6.3	8.0	11.0	11.0	12.7
T-Bill	336	243	236	88	186
Bonds	70	348	673	894	1,023
Commercial Banks	12	3	3	1	0
Arrears (suppliers)	28	28	0	0	0

Sources: Malian authorities, IMF Staff estimates.

UNDERLYING MACROECONOMIC ASSUMPTIONS AND COUNTRY CLASSIFICATION

A. Background on Macroeconomic Forecasts

4. This DSA is consistent with the macroeconomic framework underlying the Staff Report prepared for the 2019 request for the ECF-Supported Program (Box 1).

Text Table 4. Mali: Evolution of Selected Projected Macroeconomic Indicators

	2016	2017	2018	2019	2020	2021	2022	2023	Long term [1]
				Projections					
Real GDP growth									
Current DSA	5.8	5.4	4.7	5.0	5.0	4.9	4.9	4.8	4.8
Previous DSA	5.8	5.3	5.0	4.7	4.7	4.7	4.7	4.7	4.7
GDP Deflator in US dollar (percent)									
Current DSA	1.1	4.0	6.8	-1.5	2.9	2.5	2.7	2.5	2.1
Previous DSA	1.2	3.6	11.3	2.5	3.0	2.7	2.0	2.2	2.1
Overall fiscal balance (including grants, percent of GDP)									
Current DSA	-3.9	-2.9	-4.7	-3.0	-3.0	-3.0	-3.0	-3.0	-3.0
Previous DSA	-3.9	-2.9	-3.3	-3.0	-3.0	-3.0	-3.0	-3.0	-3.0
Export of goods and services growth (percent)									
Current DSA	4.3	3.8	18.8	0.1	2.5	-0.1	4.1	4.1	4.1
Previous DSA	4.3	8.0	14.4	-0.3	5.2	5.0	4.7	5.0	5.3
Import of goods and services growth (percent)									
Current DSA	9.0	-2.4	6.0	6.9	6.3	4.7	4.8	4.9	5.5
Previous DSA	9.0	5.4	14.2	3.6	5.0	5.4	5.7	-7.5	5.1
Current account deficit (including grants, percent of GDP)									
Current DSA	-7.2	-8.1	-3.8	-5.6	-6.4	-7.1	-7.0	-6.9	-6.1
Previous DSA	-7.2	-6.0	-6.5	-6.3	-6.1	-6.0	-6.3	-6.4	-6.5
Official aid [2] (percent of GDP)									
Current DSA	3.4	3.9	2.2	4.2	3.2	2.8	2.9	2.7	2.1
Previous DSA	3.4	3.9	3.4	3.8	4.2	4.4	4.6	4.4	4.5

Sources: Malian authorities; and IMF Staff estimates and projections.

[1] Defined as the last 15 years of the projection period. For the current DSA, the long term covers the 2023-37 period. For the previous DSA, it covered 2022-36.

[2] Defined as the sum of concessional grants and loans.

5. Macroeconomic conditions have been updated since the previous 2018 DSA (Text Table 4). In the short term, the main changes in assumptions relate to fiscal balance, trade and external support (Text Table 5). In the medium and long terms, key macroeconomic assumptions are in line with the previous DSA.

Box 1. Baseline Macroeconomic Assumptions

- *Real GDP growth*. The economic outlook, while presenting some encouraging elements, is subject to important downside risks. In 2019, real output growth is projected at 5.0 percent and remains near potential growth (4.8 percent) over the medium term (Text Table 4). Downside risks relate to the possible deterioration in security situation, potential terms of trade shocks (to the price of gold, cotton, and fuels), and adverse weather conditions. A specific risk arises from a continued shortfall in domestically-financed public investment because of a failure to mobilize sufficient domestic revenues. This outlook is broadly similar to the projected growth path in the previous DSA.
- *Fiscal policy*. In 2018, fiscal revenue sharply declined by 4.2 percentage points of GDP, leading to a widening of the overall deficit to 4.7 percent of GDP (from 2.9 percent of GDP in 2017). This poor performance was explained by a combination of weakened compliance and enforcement in the run-up to presidential elections, worsened security conditions, and higher international fuel prices. The authorities partially offset the revenue shortfall by cutting domestically-financed capital spending (by 1.8 percent of GDP), as well as current spending (by 1.2 percent of GDP). Staff expect revenue to recover by 2020 with a frontloading of 75 percent of the recovery in 2019. This path would help the authorities achieve an overall fiscal balance of 3 percent of GDP already in 2019, in line with the WAEMU convergence criterion.
- *External sector*. The current account deficit is projected at 5.6 percent of GDP in 2019 (6.5 in the previous DSA) from 3.8 percent of GDP in 2018, driven by imports for mining and infrastructure investment and a slowdown in gold exports. Thereafter, the current account deficit is projected to further widen to 7.1 percent of GDP in 2020 as terms of trade deteriorate, and then gradually narrow to stabilize at about 6.9 percent by 2023. This stabilization in the external position would be driven partly by supportive macroeconomic policies, gradual increase in exports (including food, cotton, and other minerals), and lower long-run oil prices. These factors should help to offset the expected steady decline in export earnings from gold.[1] As in the previous DSA, the current account deficit continues to be financed mainly through FDI, public sector borrowing, and official grants.
- *Financing mix*. External borrowing is expected to increase to 3.5 percent of GDP in 2019 from 2.2 percent of GDP in 2018, reflecting the large increase in the 2019 World Bank's budget support (from US$60 million to US$250 million). The 2019 baseline also includes the disbursement of a near-concessional loan of US$250 million from Abu Dhabi for balance of payments purposes. The authorities initially intended to use the policy-based guarantee instrument of the World Bank to contract syndicated loans in the international market for about US$500 million on commercial terms. However, there remains considerable uncertainty about the timing and size of this operation and it has not been included in the baseline for the new program. Over the medium term, gross financing needs will continue to be financed by external multilateral sources and bonds issued in the regional bond market (UEMOA-Titres).

[1] Gold export volumes are expected to decline steadily over time, with the share of gold in total exports projected to fall from 67 percent in 2015 to about 20 percent in 2036.

Text Table 5. External Financing Assumptions, 2019–30
(in percent of GDP)

	2019	2020	2021	2022	2023	2024-2029	2024	2025	2026	2027	2028	2029	2030
Total external	**5.8**	**4.4**	**4.0**	**4.1**	**3.8**	**3.7**	**3.4**	**3.3**	**3.2**	**3.2**	**3.2**	**3.1**	**3.1**
Grants	**2.3**	**2.0**	**1.7**	**1.8**	**1.8**	**1.6**	**1.3**	**1.3**	**1.3**	**1.3**	**1.2**	**1.2**	**1.2**
Projects	1.3	0.9	0.9	1.0	0.9	0.8	0.9	0.9	0.8	0.8	0.8	0.7	0.7
Budget support	1.2	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
External loans	**3.5**	**2.3**	**2.3**	**2.3**	**2.1**	**2.1**	**2.1**	**2.1**	**2.1**	**2.1**	**2.1**	**2.1**	**2.1**
Concessional loan of which [1]	**1.9**	**1.2**	**1.1**	**1.0**	**1.0**	**0.8**	**0.9**	**0.8**	**0.8**	**0.8**	**0.7**	**0.7**	**0.7**
Multilateral of which	1.5	0.9	0.8	0.8	0.7	0.6	0.6	0.6	0.5	0.5	0.5	0.4	0.4
World Bank	1.1	0.5	0.4	0.4	0.4	0.3	0.3	0.3	0.3	0.3	0.3	0.2	0.2
Bilateral	0.4	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.2	0.2
Nonconcessional loan [2]	**1.6**	**1.2**	**1.2**	**1.3**	**1.1**	**1.2**	**1.2**	**1.2**	**1.2**	**1.3**	**1.3**	**1.4**	**1.4**
Of which: IMF financing	0.2	0.3	0.3	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Sources: Malian authorities and staff estimates.
[1] Debt in which the grant element exceeds 35 percent.
[2] Debt comprising a grant element that is positive but is less than 35 percent.

6. The realism tools show that projections are broadly in line with historical and peers' experiences (Figure 6)

- **Forecast errors**. Projections of the drivers of external debt are in line with past debt dynamics, when unexpected changes in debt were mainly driven by unexpected changes in the current account and FDI. On average, primary deficit was the main contributor of total public debt.

- **Fiscal adjustment**. The fiscal adjustment path is in line with peer country experiences but with a lower impact on medium-term GDP growth than suggested by fiscal multipliers. The primary sector and gold production, which are weakly related to the fiscal stance, account for about 40 percent of real GDP growth. Moreover, the fiscal consolidation in 2020 is mainly driven by an increase in revenue (smaller fiscal multiplier) rather than a decrease in public expenditure (higher fiscal multiplier).[2]

- **Investment-growth**. The contribution of government capital stock to growth is similar to that observed in the previous DSA.

B. Country Classification and Determination of Scenario Stress Tests

7. Mali is assessed as having a medium debt carrying capacity. Based on the April 2019 WEO macroeconomic framework and updated World Bank's CPIA measures, Mali's composite indicator score lies between 2.69 and 3.05, in line with the assessment of medium debt carrying capacity applied in the May 2018 DSA.

[2] See the Sub-Saharan Africa Fiscal Adjustment and Economic Diversification, Regional Economic Outlook, IMF, October 2017 for more details on fiscal multipliers in Sub-Saharan African countries.

Applicable Thresholds

APPLICABLE	
EXTERNAL debt burden thresholds	
PV of debt in % of	
Exports	180
GDP	40
Debt service in % of	
Exports	15
Revenue	18

APPLICABLE	
TOTAL public debt benchmark	
PV of total public debt in percent of GDP	55

Calculation of the CI Index

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	3.389	1.30	43%
Real growth rate (in percent)	2.719	5.361	0.15	5%
Import coverage of reserves (in percent)	4.052	39.075	1.58	53%
Import coverage of reserves^2 (in percent)	-3.990	15.269	-0.61	-20%
Remittances (in percent)	2.022	4.649	0.09	3%
World economic growth (in percent)	13.520	3.559	0.48	16%
CI Score			3.00	100%
CI rating			Medium	

Debt Carrying Capacity	Medium

Final	Classification based on current vintage	Classification based on the previous vintage	Classification based on the two previous vintages
Medium	Medium 3.00	Medium 3.01	Medium 2.99

Note: Until the April 2019 WEO vintage is released, the two previous vintages ago classification and corresponding score are based solely on the CPIA per the previous framework.

C. External DSA

8. The risk of external debt distress for Mali remains moderate with some space to absorb shocks. All debt indicators remain below their corresponding indicative thresholds under the baseline scenario and stress tests, indicating space to absorb shocks; however, in a customized alternative scenario in which the fiscal deficit is 2 percentage points of GDP per year higher than in the baseline during 2020-23, the debt service to export ratio breaches the threshold (Figure 4).

9. Mali's external debt ratios remain low under the baseline scenario. Under the baseline scenario, the ratio for the PV of external public debt to GDP is projected to remain between 14 and 17 percent of GDP, well below the indicative threshold of 40 percent throughout the projection period (Figure 1, Table 1). The PV of the external debt-to-exports is also projected to remain broadly stable between 73.0 percent and 118.0 percent, comfortably below the 180 percent threshold. Debt service indicators are all below their corresponding thresholds. Including the PBG and reprofiling operation would not change the risk rating. The standardized stress tests show a negative impact on the trajectory of debt indicators, yet no threshold is breached.

10. However, security challenges and large development needs justify considering a customized scenario with higher deficits which underscores the vulnerability of Mali's external debt to external shocks (Figure 4). This scenario would reflect additional security-related expenditures that may be required if the conflict intensifies and the large investment spending projected in the authorities' CREDD. The deficit would remain above the WAEMU's convergence criterion of 3 percent of GDP by 2 percentage points of GDP for the next four years and financed on commercial terms.[3] Debt ratios significantly would increase compared to the baseline, leading to a breach of the debt service-to-exports ratio under an export shock (the most extreme).

11. Mali's external debt sustainability is highly sensitive to an export growth shock, a reduction in current transfers and FDI and exchange rate depreciations, along with changes in borrowing terms. Under a bound test with a temporary reduction in export growth[4] in 2020–21, the debt service-to-exports ratio significantly increases under the baseline scenario and would breach its threshold under the customized scenario in 2026-27 (Figure 4). Similarly, a bound test with lower FDI and current transfers[5] in 2020–21 would drive the PV of the debt-to-exports ratio toward the threshold.

12. Going forward, macroeconomic stability is crucial to maintaining external debt sustainability. In particular, it is essential that the authorities continue their efforts to mobilize domestic revenue and implement reforms to diversify the exports base. The full implementation of the 2015 peace agreement is also critical to macroeconomic stability and debt sustainability.

D. Public DSA

13. The inclusion of domestic debt does not alter the assessment of Mali's debt sustainability. Given the small size of Mali's domestic debt and the planned domestic debt reprofiling operation, the public debt sustainability analysis closely mirrors the external debt sustainability analysis under both the baseline and alternative scenarios (Figures 4 and 5). The PV of public sector debt-to-GDP ratio stays between 30.6 and 34.4 percent of GDP during the entire projection period under the baseline scenario but increases to 39.1 percent of GDP in 2023 under the alternative scenario of higher deficits financed at less

[3] The average grant element is 0.85 percent with an unified discount rate of 5 percent.

[4] Nominal export growth (in USD) set to its historical average minus one standard deviation, or the baseline projection minus one standard deviation, whichever is lower, in the second and third years of the projection period.

[5] Current transfers-to-GDP and FDI-to-GDP ratios set to their historical average minus one standard deviation, or baseline projection minus one standard deviation, whichever is lower, in the second and third years of the projection period.

favorable terms. That said, the recent rapid growth of domestic debt stock needs to be monitored closely to maintain debt sustainability and financial stability going forward.

E. Conclusion

14. Mali's risk of external debt distress and overall risk of debt distress remain moderate. As in the previous DSA, while Mali's public debt remains low, debt sustainability is sensitive to external shocks. All debt indicators remain below their corresponding indicative thresholds under the baseline scenario and stress tests. However, an export shock a customized scenario of higher deficits financed at unfavorable terms shows a breach of the debt service-to-exports threshold. The customized scenario reflects additional security-related expenditures that may be required if the conflict intensifies. Mali's debt sustainability is also highly sensitive to a tightening of financing terms, to an exports shock given the export concentration in gold and cotton, and to a reduction in current transfers and FDI. In addition, non-resident holdings of securities issued in the regional securities market can pose additional risks that are not currently captured in the external DSA, although the scale of such holdings is unclear. Mali needs to maintain prudent macroeconomic policies, strengthen the effectiveness of public investment, debt management and continue to meet its external financing needs with grants and concessional loans, wherever possible. In addition, projects should deliver high returns; and structural reforms to improve the investment climate and export diversification should continue, in view of an expected decline in gold's exports over the medium term.

F. Authorities' Views

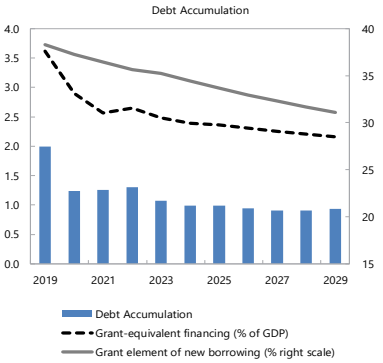
15. The authorities concurred with the results of the DSA. They agree on the need to tackle the country's debt sustainability challenges. They commit to seeking grants and concessional financing, avoiding as much as possible non-concessional borrowing.

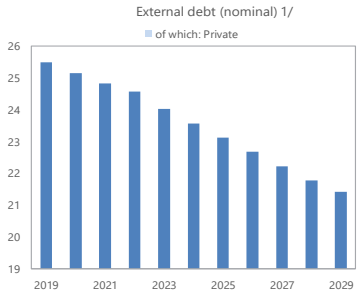
Table 1. Mali: External Debt Sustainability Framework, Baseline Scenario, 2016–39

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
External debt (nominal) 1/	**25.1**	**24.9**	**24.6**	**25.5**	**25.1**	**24.8**	**24.6**	**24.0**	**23.6**	**21.4**	**16.8**	**22.3**	**23.5**
of which: public and publicly guaranteed (PPG)	**25.1**	**24.9**	**24.6**	**25.5**	**25.1**	**24.8**	**24.6**	**24.0**	**23.6**	**21.4**	**16.8**	**22.3**	**23.5**
Change in external debt	2.4	-0.1	-0.3	0.9	-0.4	-0.3	-0.3	-0.5	-0.5	-0.4	-2.3		
Identified net debt-creating flows	**3.9**	**1.5**	**-1.9**	**1.9**	**2.9**	**3.9**	**4.0**	**3.9**	**3.9**	**3.0**	**1.9**	**1.7**	**3.4**
Non-interest current account deficit	**7.0**	**7.0**	**3.5**	**5.3**	**6.2**	**6.9**	**6.8**	**6.7**	**6.6**	**5.7**	**4.4**	**5.9**	**6.3**
Deficit in balance of goods and services	16.9	13.7	10.5	12.3	12.9	13.6	13.3	13.2	13.2	12.8	13.4	**12.6**	**13.0**
Exports	23.5	22.2	23.6	22.8	21.7	20.1	19.4	18.8	18.3	15.9	12.1		
Imports	40.3	35.9	34.1	35.2	34.6	33.7	32.8	32.0	31.5	28.7	25.5		
Net current transfers (negative = inflow)	-12.3	-9.7	-9.4	-9.4	-9.0	-8.8	-8.5	-8.4	-8.4	-8.4	-6.0	**-9.9**	**-8.6**
of which: official	-7.4	-4.8	-4.4	-4.4	-4.0	-3.8	-3.6	-3.4	-3.4	-3.3	-0.9		
Other current account flows (negative = net inflow)	2.4	3.0	2.4	2.4	2.2	2.1	2.0	1.9	1.7	1.2	-3.1	**3.2**	**1.8**
Net FDI (negative = inflow)	**-1.9**	**-3.6**	**-3.0**	**-2.5**	**-2.4**	**-2.1**	**-1.9**	**-1.9**	**-1.9**	**-1.9**	**-1.9**	**-3.2**	**-2.0**
Endogenous debt dynamics 2/	**-1.2**	**-1.9**	**-2.4**	**-1.0**	**-0.9**	**-0.9**	**-0.9**	**-0.9**	**-0.8**	**-0.7**	**-0.6**		
Contribution from nominal interest rate	0.3	0.3	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.3	0.3		
Contribution from real GDP growth	-1.2	-1.2	-1.0	-1.2	-1.2	-1.1	-1.1	-1.1	-1.1	-1.0	-0.9		
Contribution from price and exchange rate changes	-0.2	-1.0	-1.6	...	...	...	...	...	...	...	...		
Residual 3/	**-1.5**	**-1.6**	**1.5**	**-1.0**	**-3.2**	**-4.2**	**-4.2**	**-4.4**	**-4.3**	**-3.4**	**-4.2**	**-1.1**	**-3.7**
of which: exceptional financing	-0.2	-0.2	-0.2	-0.2	-0.1	-0.1	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**15.3**	**16.7**	**16.6**	**16.6**	**16.6**	**16.4**	**16.3**	**15.4**	**13.1**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**64.6**	**73.0**	**76.5**	**82.3**	**85.3**	**87.3**	**88.9**	**97.1**	**108.5**		
PPG debt service-to-exports ratio	**4.6**	**4.7**	**4.0**	**4.1**	**5.1**	**5.4**	**5.4**	**5.9**	**6.5**	**8.0**	**10.9**		
PPG debt service-to-revenue ratio	**7.0**	**6.3**	**7.7**	**5.9**	**6.6**	**6.4**	**6.0**	**6.3**	**6.7**	**6.7**	**...**		
Gross external financing need (Million of U.S. dollars)	868.1	690.6	254.2	675.4	948.3	1219.7	1312.0	1401.9	1510.6	1801.7	2665.5		
Key macroeconomic assumptions													
Real GDP growth (in percent)	5.8	5.4	4.7	5.0	5.0	4.9	4.9	4.8	4.8	4.8	4.8	**4.4**	**4.9**
GDP deflator in US dollar terms (change in percent)	1.1	4.0	6.8	-1.5	2.9	2.5	2.7	2.5	2.1	2.1	2.1	**1.6**	**2.0**
Effective interest rate (percent) 4/	1.3	1.1	1.0	0.9	1.0	1.0	1.0	1.1	1.1	1.3	1.6	**0.4**	**1.1**
Growth of exports of G&S (US dollar terms, in percent)	4.3	3.8	18.8	0.1	2.5	-0.1	4.1	4.1	4.0	4.0	4.1	**5.3**	**3.2**
Growth of imports of G&S (US dollar terms, in percent)	9.0	-2.4	6.0	6.9	6.3	4.7	4.8	4.9	5.5	5.0	5.9	**4.6**	**5.3**
Grant element of new public sector borrowing (in percent)	...	...	...	38.3	37.3	36.5	35.7	35.2	34.4	31.1	#DIV/0!	**...**	**34.5**
Government revenues (excluding grants, in percent of GDP)	15.5	16.6	12.4	16.0	16.7	16.9	17.3	17.6	17.8	18.8	0.0	**13.8**	**17.7**
Aid flows (in Million of US dollars) 5/	223.6	248.1	210.8	747.2	613.8	584.3	639.4	657.7	672.2	807.4	0.0		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	3.6	2.9	2.6	2.6	2.5	2.4	2.2	...	**...**	**2.5**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	62.9	66.4	63.5	64.1	65.1	64.0	60.4	...	**...**	**63.2**
Nominal GDP (Million of US dollars)	14,016	15,368	17,179	17,780	19,217	20,669	22,270	23,927	25,602	35,910	70,647		
Nominal dollar GDP growth	6.9	9.6	11.8	3.5	8.1	7.6	7.7	7.4	7.0	7.0	7.0	**6.1**	**6.9**
Memorandum items:													
PV of external debt 7/	...	...	15.3	16.7	16.6	16.6	16.6	16.4	16.3	15.4	13.1		
In percent of exports	...	...	64.6	73.0	76.5	82.3	85.3	87.3	88.9	97.1	108.5		
Total external debt service-to-exports ratio	4.6	4.7	4.0	4.1	5.1	5.4	5.4	5.9	6.5	8.0	10.9		
PV of PPG external debt (in Million of US dollars)			2621.0	2963.3	3183.4	3424.2	3693.4	3931.4	4168.2	5547.3	9244.0		
(PVt-PVt-1)/GDPt-1 (in percent)				2.0	1.2	1.3	1.3	1.1	1.0	0.9	-0.7		
Non-interest current account deficit that stabilizes debt ratio	4.6	7.1	3.9	4.5	6.6	7.2	7.0	7.2	7.1	6.0	6.6		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

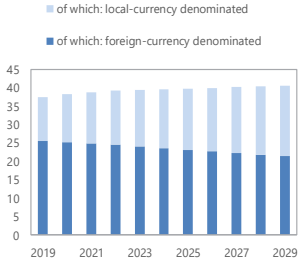
Table 2. Mali: Public Sector Debt Sustainability Framework, Baseline Scenario, 2016–2039

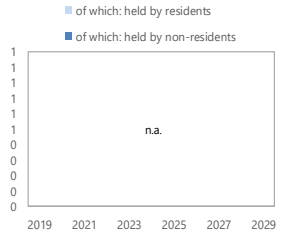
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
Public sector debt 1/	**36.0**	**36.0**	**37.3**	**37.4**	**38.2**	**38.7**	**39.2**	**39.3**	**39.6**	**40.6**	**43.6**	**29.0**	**39.4**
of which: external debt	**25.1**	**24.9**	**24.6**	**25.5**	**25.1**	**24.8**	**24.6**	**24.0**	**23.6**	**21.4**	**16.8**	**22.3**	**23.5**
Change in public sector debt	5.4	-0.1	1.3	0.1	0.8	0.5	0.5	0.1	0.2	0.2	-1.0		
Identified debt-creating flows	**2.8**	**-2.0**	**3.4**	**0.1**	**0.6**	**0.4**	**0.3**	**0.3**	**0.4**	**0.4**	**-1.0**	**0.8**	**0.4**
Primary deficit	**3.4**	**2.1**	**3.9**	**2.1**	**2.0**	**1.9**	**1.9**	**1.8**	**1.8**	**1.5**	**0.0**	**2.3**	**1.8**
Revenue and grants	17.1	18.2	13.6	18.3	18.7	18.7	19.1	19.3	19.5	20.4	0.0	**16.0**	**19.4**
of which: grants	1.6	1.6	1.2	2.3	2.0	1.7	1.8	1.8	1.7	1.5	0.0		
Primary (noninterest) expenditure	20.4	20.3	17.5	20.4	20.7	20.6	21.0	21.1	21.2	21.9	0.0	**18.3**	**21.2**
Automatic debt dynamics	**-0.7**	**-4.3**	**-0.6**	**-1.8**	**-1.6**	**-1.6**	**-1.5**	**-1.5**	**-1.3**	**-1.2**	**-1.0**		
Contribution from interest rate/growth differential	-1.4	-1.7	-1.5	-1.5	-1.5	-1.4	-1.4	-1.4	-1.3	-1.2	-1.0		
of which: contribution from average real interest rate	0.3	0.2	0.1	0.3	0.3	0.4	0.4	0.4	0.5	0.6	1.0		
of which: contribution from real GDP growth	-1.7	-1.9	-1.6	-1.8	-1.8	-1.8	-1.8	-1.8	-1.8	-1.8	-2.0		
Contribution from real exchange rate depreciation	0.6	-2.6	1.0	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.2**	**0.1**	**0.1**	**-0.3**	**0.2**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**-0.4**	**0.0**
Privatization receipts (negative)	0.2	0.1	0.1	-0.3	0.2	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**2.5**	**2.0**	**-2.1**	**-0.2**	**0.1**	**0.0**	**0.0**	**-0.3**	**-0.2**	**-0.1**	**0.0**	**0.9**	**-0.1**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**28.5**	**28.5**	**29.6**	**30.4**	**31.2**	**31.7**	**32.2**	**34.6**	**39.9**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**209.4**	**155.4**	**158.2**	**162.8**	**162.9**	**163.9**	**165.4**	**169.9**	**...**		
Debt service-to-revenue and grants ratio 3/	**32.8**	**35.1**	**37.6**	**26.5**	**22.9**	**30.1**	**30.5**	**33.3**	**34.9**	**39.5**	**...**		
Gross financing need 4/	9.2	8.6	9.1	6.7	6.5	7.5	7.7	8.3	8.6	9.6	10.7		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	5.8	5.4	4.7	5.0	5.0	4.9	4.9	4.8	4.8	4.8	4.8	**4.4**	**4.9**
Average nominal interest rate on external debt (in percent)	1.3	1.2	1.0	0.9	1.0	1.0	1.0	1.1	1.1	1.3	1.6	**0.4**	**1.1**
Average real interest rate on domestic debt (in percent)	3.4	3.1	3.3	3.9	5.0	4.9	4.8	4.7	4.7	4.7	4.7	**2.5**	**4.7**
Real exchange rate depreciation (in percent, + indicates depreciation)	2.9	-11.0	4.2	...	...	...	...	...	...	...	...	**0.2**	**...**
Inflation rate (GDP deflator, in percent)	1.4	1.9	2.1	1.8	1.8	2.0	2.1	2.1	2.1	2.1	2.1	**3.7**	**2.0**
Growth of real primary spending (deflated by GDP deflator, in percent)	16.1	4.9	-9.8	22.4	6.5	4.2	7.0	5.6	5.3	5.5	-100.0	**6.2**	**7.1**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-2.0	2.2	2.6	2.0	1.2	1.4	1.4	1.7	1.5	1.3	1.0	**0.9**	**1.5**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government . Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Mali: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2019–29

Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	No	
Natural disaster	n.a.	n.a.
Commodity price	No	No
Market financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

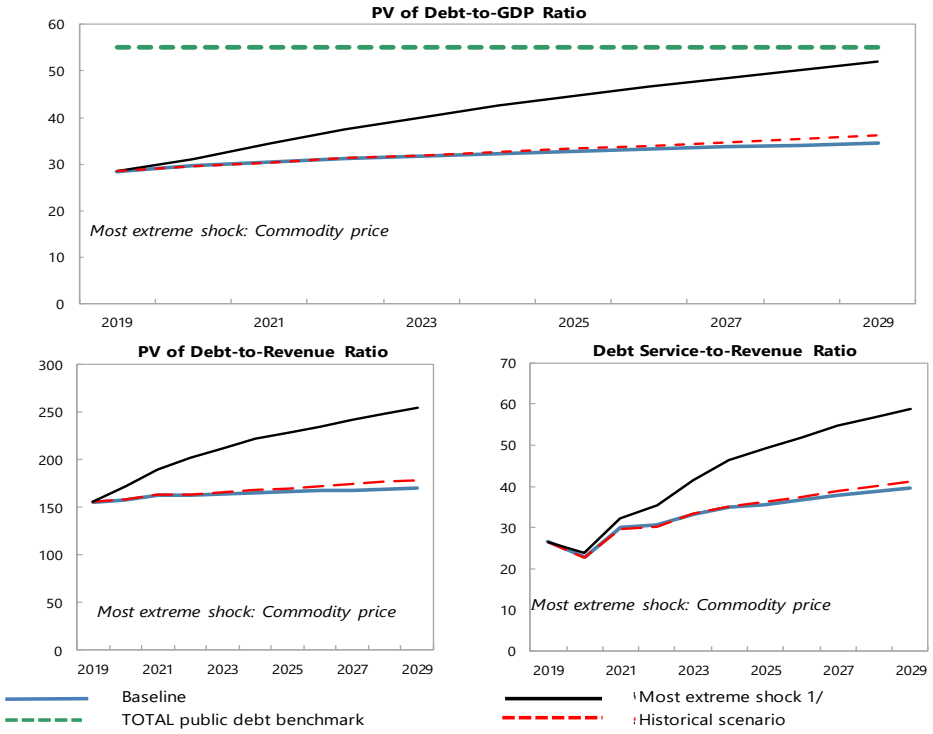
Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.5%	1.5%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	23	23
Avg. grace period	5	5

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Figure 2. Mali: Indicators of Public Debt Under Alternative Scenarios, 2019–29



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	28%	28%
Domestic medium and long-term	65%	65%
Domestic short-term	7%	7%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.5%	1.5%
Avg. maturity (incl. grace period)	23	23
Avg. grace period	5	5
Domestic MLT debt		
Avg. real interest rate on new borrowing	4.8%	4.8%
Avg. maturity (incl. grace period)	4	4
Avg. grace period	1	1
Domestic short-term debt		
Avg. real interest rate	4.5%	4.5%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Mali: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2019–2029

(In percent)

	Projections 1/										
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of debt-to GDP ratio											
Baseline	19	19	18	18	17	17	16	16	16	15	15
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	19	18	18	17	16	16	15	15	15	14	14
B. Bound Tests											
B1. Real GDP growth	19	19	20	20	19	18	18	17	17	17	16
B2. Primary balance	19	19	19	19	18	18	18	17	17	16	16
B3. Exports	19	20	22	22	21	21	20	19	19	18	18
B4. Other flows 3/	19	22	25	25	24	23	23	22	21	20	19
B5. Depreciation	19	23	18	18	18	17	17	16	16	16	16
B6. Combination of B1-B5	19	23	24	23	22	22	21	21	20	19	19
C. Tailored Tests											
C1. Combined contingent liabilities	19	20	20	20	20	20	19	19	19	19	18
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	19	19	19	19	18	18	17	16	16	15	15
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	85	88	90	90	89	89	88	87	85	84	84
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	85	86	86	85	84	83	81	80	80	80	80
B. Bound Tests											
B1. Real GDP growth	85	88	90	90	89	89	88	87	85	84	84
B2. Primary balance	85	89	93	94	93	94	93	92	91	91	90
B3. Exports	85	103	130	129	128	128	126	124	121	119	117
B4. Other flows 3/	85	106	125	124	123	123	121	118	115	112	109
B5. Depreciation	85	88	72	72	72	71	70	69	69	70	70
B6. Combination of B1-B5	85	107	106	115	114	114	112	110	108	105	104
C. Tailored Tests											
C1. Combined contingent liabilities	85	96	99	100	102	103	103	103	103	103	103
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	85	93	97	97	95	94	91	89	87	85	84
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	4	6	6	7	8	8	8	8	8	8	8
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	4	6	6	7	8	8	8	8	8	8	8
B. Bound Tests											
B1. Real GDP growth	4	6	6	7	8	8	8	8	8	8	8
B2. Primary balance	4	6	6	7	8	8	8	8	9	9	8
B3. Exports	4	6	7	9	10	10	10	11	12	12	11
B4. Other flows 3/	4	6	6	8	8	9	9	10	11	11	11
B5. Depreciation	4	6	6	7	7	7	7	8	7	7	7
B6. Combination of B1-B5	4	6	7	8	9	9	9	10	10	10	10
C. Tailored Tests											
C1. Combined contingent liabilities	4	6	6	7	8	8	8	8	9	9	9
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	4	6	6	8	8	8	8	8	9	9	8
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	6	7	7	8	8	8	8	8	8	7	7
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	6	7	7	8	8	8	8	8	8	7	7
B. Bound Tests											
B1. Real GDP growth	6	7	7	9	9	9	8	8	8	8	8
B2. Primary balance	6	7	7	8	8	8	8	8	8	8	7
B3. Exports	6	7	7	8	8	9	8	9	9	9	8
B4. Other flows 3/	6	7	7	9	9	9	8	9	10	10	9
B5. Depreciation	6	9	8	9	9	9	9	9	8	8	7
B6. Combination of B1-B5	6	7	8	9	9	9	9	10	10	9	9
C. Tailored Tests											
C1. Combined contingent liabilities	6	7	7	8	8	8	8	8	8	8	7
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	6	7	7	8	8	8	8	8	8	8	7
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Mali: Sensitivity Analysis for Key Indicators of Public Debt, 2019–2029

	Projections 1/										
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of Debt-to-GDP Ratio											
Baseline	30	31	32	33	33	33	34	34	35	35	35
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	30	31	32	33	33	34	35	35	36	37	37
B. Bound Tests											
B1. Real GDP growth	30	33	37	39	41	43	45	46	48	50	51
B2. Primary balance	30	33	35	35	36	36	36	36	37	37	37
B3. Exports	30	33	36	36	36	37	37	37	37	37	38
B4. Other flows 3/	30	35	39	39	40	40	40	40	40	40	40
B5. Depreciation	30	35	34	33	32	31	30	29	29	28	27
B6. Combination of B1-B5	30	31	33	32	33	33	33	33	34	34	35
C. Tailored Tests											
C1. Combined contingent liabilities	30	39	39	40	40	40	40	41	41	41	41
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	30	33	36	39	41	44	46	48	49	51	53
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
TOTAL public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	167	165	166	165	165	165	165	165	165	165	165
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	167	165	166	167	167	169	170	172	173	175	176
B. Bound Tests											
B1. Real GDP growth	167	175	190	197	203	210	216	223	228	234	239
B2. Primary balance	167	172	179	178	177	177	177	176	176	175	175
B3. Exports	167	172	184	183	182	181	181	180	179	177	176
B4. Other flows 3/	167	185	202	200	198	197	196	194	191	188	186
B5. Depreciation	167	186	176	168	161	155	149	143	137	132	127
B6. Combination of B1-B5	167	166	169	165	164	163	163	162	162	162	162
C. Tailored Tests											
C1. Combined contingent liabilities	167	205	204	202	199	198	197	196	195	193	192
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	167	179	192	204	212	219	225	230	236	241	247
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	27	23	29	31	34	35	36	38	39	40	40
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	27	23	29	31	34	36	37	38	40	41	42
B. Bound Tests											
B1. Real GDP growth	27	24	32	35	40	44	47	50	53	55	57
B2. Primary balance	27	23	30	34	37	39	40	40	41	42	43
B3. Exports	27	23	29	31	34	36	37	38	40	41	41
B4. Other flows 3/	27	23	30	32	35	36	37	39	41	42	42
B5. Depreciation	27	23	29	31	33	34	35	37	38	38	38
B6. Combination of B1-B5	27	23	29	31	34	36	37	37	38	39	40
C. Tailored Tests											
C1. Combined contingent liabilities	27	23	34	41	44	47	43	45	46	46	45
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	27	24	31	36	42	46	50	52	55	57	59
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.

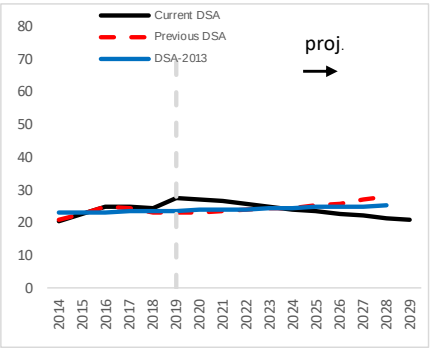
1/ A bold value indicates a breach of the benchmark.

2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

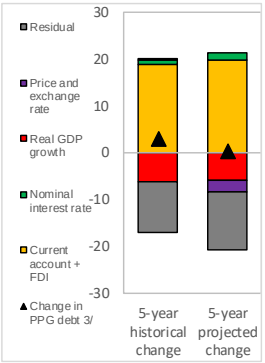
3/ Includes official and private transfers and FDI.

Figure 3. Mali: Drivers of Debt Dynamics – Baseline Scenario

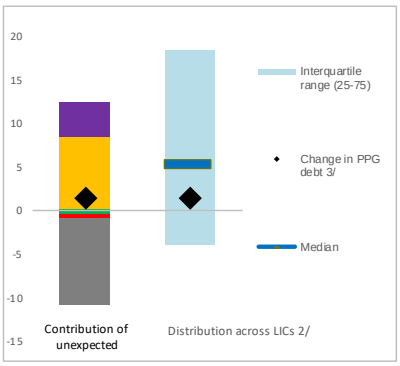
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

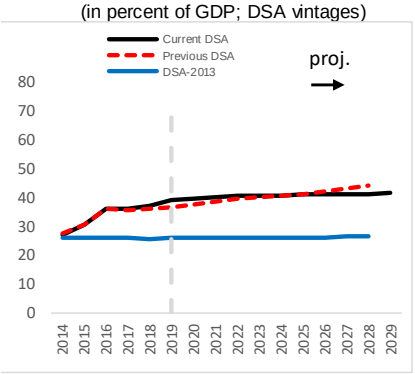


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

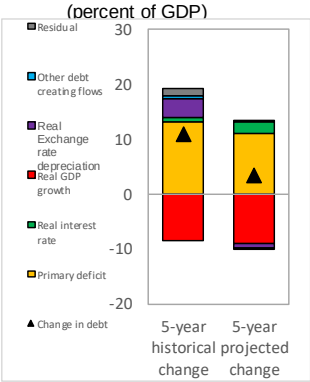


Public debt

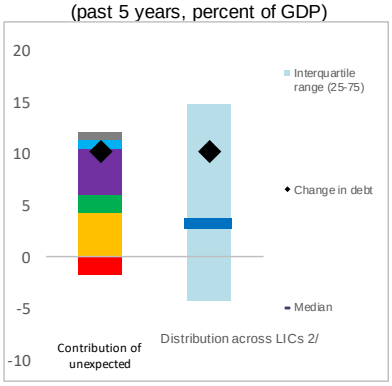
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



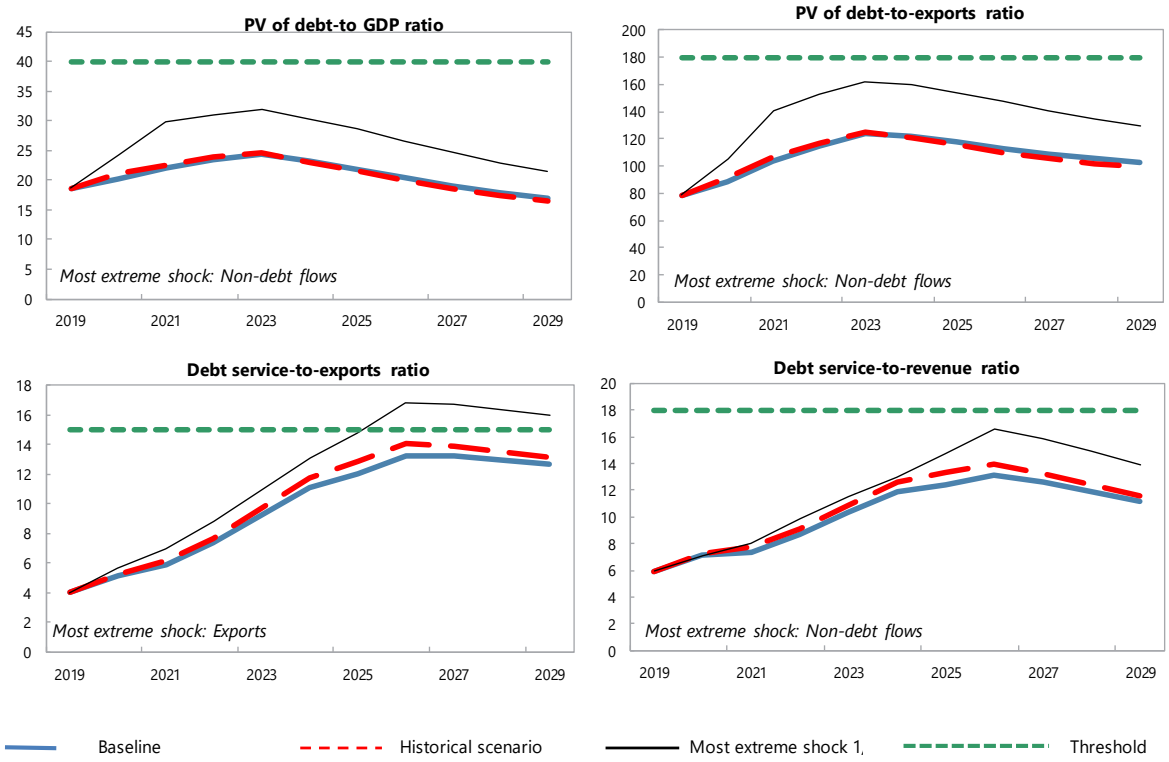
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

Sources: Country authorities and staff estimates and projections.

Figure 4. Mali: Indicators of Public and Publicly Guaranteed External Debt Under Customized Scenarios, 2019–2029



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	No	
Natural disaster	n.a.	n.a.
Commodity price	No	No
Market financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

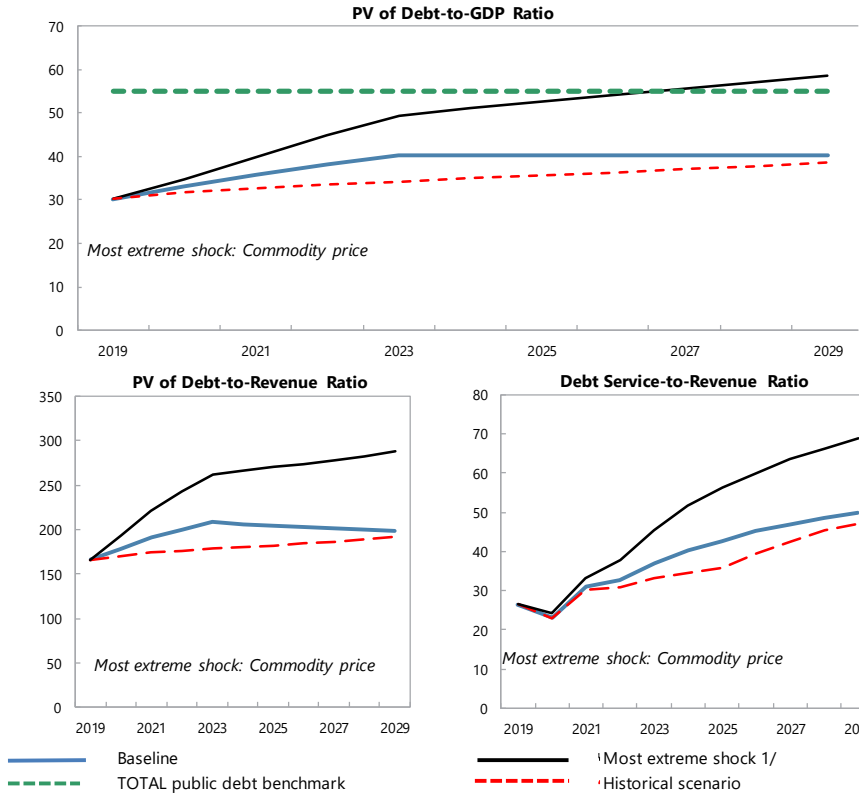
Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	2.3%	2.3%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	20	20
Avg. grace period	4	4

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Figure 5. Mali: Indicators of Public Debt Under Customized Scenarios, 2019–2029



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	33%	33%
Domestic medium and long-term	61%	61%
Domestic short-term	7%	7%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	2.3%	2.3%
Avg. maturity (incl. grace period)	20	20
Avg. grace period	4	4
Domestic MLT debt		
Avg. real interest rate on new borrowing	4.7%	4.7%
Avg. maturity (incl. grace period)	4	4
Avg. grace period	1	1
Domestic short-term debt		
Avg. real interest rate	4.4%	4.4%

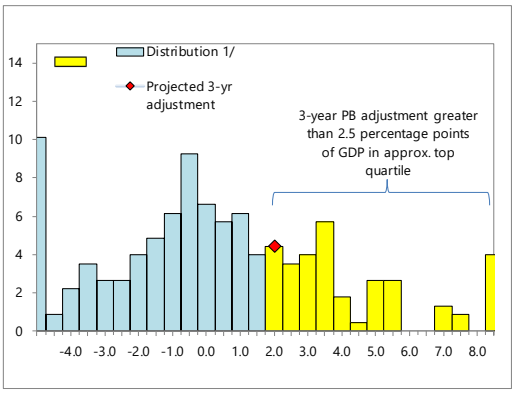
* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

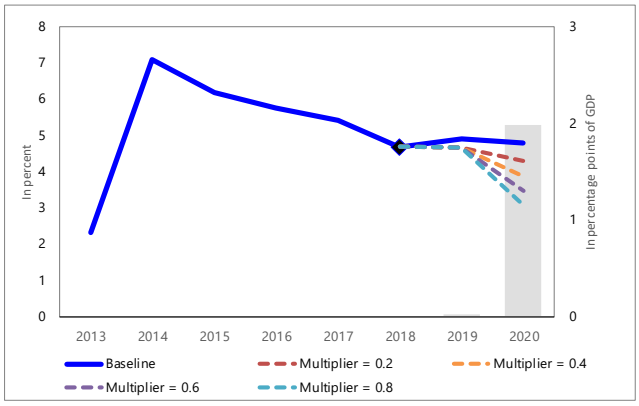
Figure 6. Mali: Realism Tools

3-Year Adjustment in Primary Balance (Percentage points of GDP)



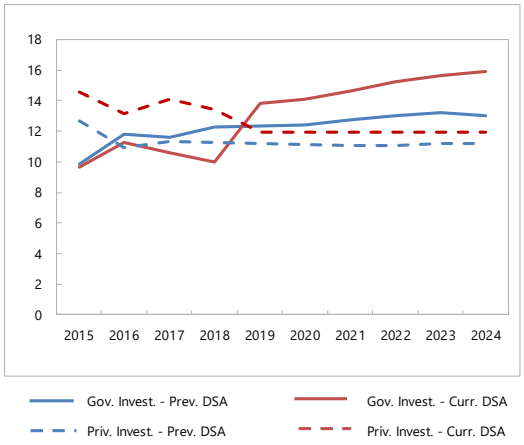
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/

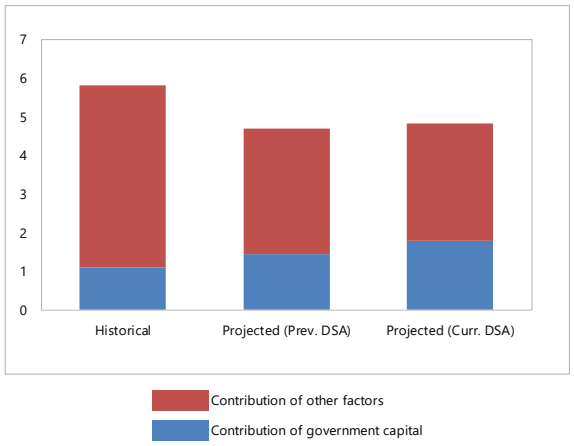


1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)

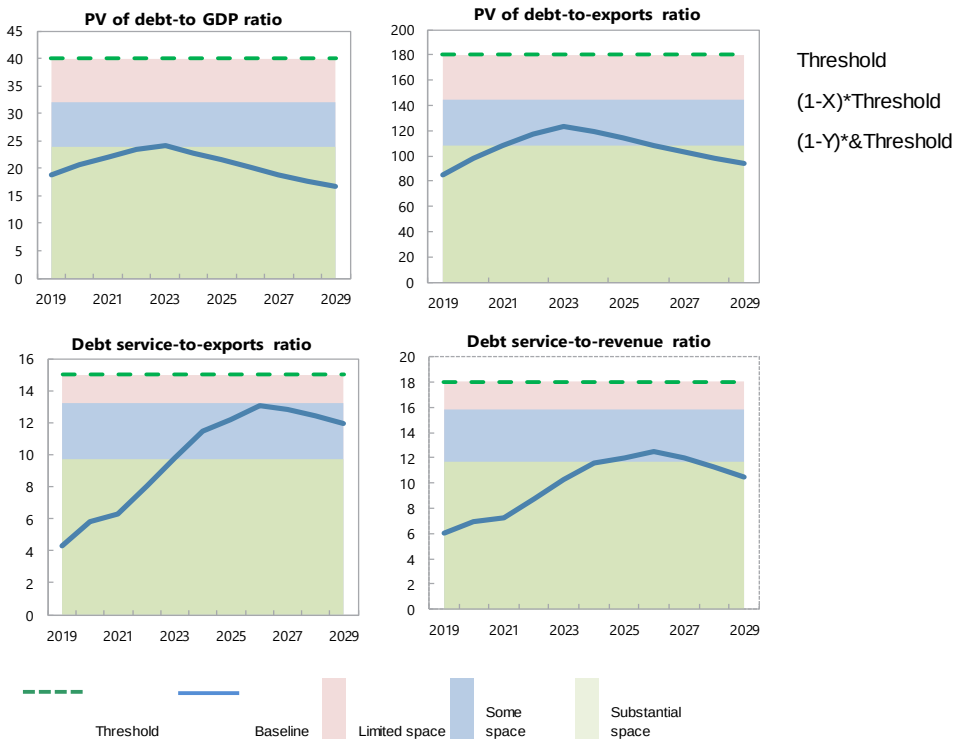


Contribution to Real GDP growth
(percent, 5-year average)



Sources: Country authorities and staff estimates and projections.

Figure 7. Mali: Qualification of the Moderate Risk Category. 2019–2029 1/



Sources: Country authorities; and staff estimates and projections.

1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.